TAYLAN ÖZGÜR ÖZDEMIR

Game Developer & Simulation Systems Engineer

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EDUCATION

İnönü University *Sep 2021 – Jun 2025*
Bachelor of Science in Computer Engineering — **GPA: 3.09 / 4.00** Malatya, Turkey
Relevant Coursework: Data Structures & Algorithms, Object-Oriented Programming, Computer Vision, Distributed Systems, Operating Systems

PROFESSIONAL EXPERIENCE

SoloRota *Jul 2025 – Jan 2026*
Game Developer (Full-time) Ankara, Turkey (On-site)

- Shipped core gameplay loops and state machines for **7 educational mobile games** to Google Play and App Store using Unity (C#).
- Architected an asynchronous data layer for REST API question streams, keeping player state intact across network drops.
- Profiled draw calls and garbage collection overhead via Unity Frame Debugger, stabilizing frame rates at **60 FPS** on mid-tier Android devices.
- Built modular in-game economies and global buff systems on `ScriptableObject` architecture, cutting cross-module dependencies.

KEY PROJECTS

Aegis Forge — Tactical Terraforming Multiplayer Shooter *github.com/taylan1477/AegisForge*
Unity, C#, DOTS, Jobs System, Burst Compiler, Native Collections, FishNet, Linux Headless

- Engineered a hybrid OOP + DOP architecture to execute real-time terrain manipulation, decoupling high-level gameplay from intensive data pipelines.
- Implemented multi-threaded height displacement and mesh modification using the **Unity C# Job System** and **Burst Compiler**, achieving sub-millisecond execution over **10,000+ dynamic tiles**.
- Built a deterministic **Event Sourcing** network replication layer deployed on a headless Linux dedicated server with FishNet, eliminating raw mesh bandwidth overhead.

SoloIdle — Educational Campus Management & Idle Game *github.com/taylan1477/SoloIdle*
Unity 2D, C#, Asynchronous Networking, JSON Serialization, ScriptableObjects

- Engineered a scalable idle economy engine powered by an asynchronous network layer and high-throughput JSON serialization for live API quiz sync.
- Structured mathematical progression algorithms, research trees, and persistence routines with `ScriptableObject` architecture to eliminate runtime data corruption.

EvoluCat — 2D Metroidvania Platformer *github.com/taylan1477/EvoluCat*
Unity 2D, C#, Finite State Machines (FSM), Physics2D, Multi-Slot Persistence

- Programmed responsive character controllers featuring wall-jump, telekinesis, and puzzle interactions via hierarchical Finite State Machines.
- Implemented an encrypted multi-slot JSON/binary save-load persistence pipeline tracking player progression across segmented world maps.

TECHNICAL SKILLS

Game & Simulation Engines: Unity Engine (2D/3D), Unity DOTS (C# Job System, Burst Compiler, NativeArray), Unreal Engine (Basics)

Programming Languages: C#, C++, Python, Java, SQL, TypeScript, Bash

Networking & Systems: FishNet, Client-Server Architecture, Linux Headless Dedicated Servers, Event Sourcing, REST APIs

Architecture & Patterns: OOP, Data-Oriented Programming (DOP), Finite State Machines (FSM), ScriptableObjects, SOLID

Developer Tools: Git, GitHub, Unity Profiler, Frame Debugger, JetBrains Rider, Visual Studio, Docker